# Brian Shu

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## Education

**Purdue University** — West Lafayette, IN
*M.S. in Computer Science* — Expected 2028

**University of Hawai'i at Mānoa** — Honolulu, HI
*B.S. in Computer Science*, GPA: 3.93/4.00 — May 2026
*B.B.A. in Finance*

**Relevant Coursework:** Machine Learning, Neural Networks & Deep Learning, Large Language Models (LLMs), Data Structures & Algorithms, Software Engineering, Databases, Data Networks
**Programming Languages:** Python, C, C++, Java, JavaScript/TypeScript, SQL
**Robotics & Simulation:** ROS 2, Gazebo, NVIDIA Isaac Lab, LeRobot
**Frameworks & Tools:** PyTorch, TensorFlow, scikit-learn, Slurm, React, Isabelle/HOL

## Publications

P. Bushuyeu*, Y. Chen*, A. Nikolaev*, **B. Shu***, and I. Molybog, "Fiatlux: A Long-Horizon Benchmark for Humanoid Ladder Climbing and Light-Bulb Replacement," *under review at IEEE ICRA 2027*. *Equal contribution | Project page

## Research Experience

**Cognitive Robot Autonomy and Learning (CoRAL) Lab, Purdue University** — West Lafayette, IN
*Graduate Researcher* (Advisor: Prof. Ahmed H. Qureshi) — Aug 2026 – Present

- Developing a **general-purpose robot manipulation policy** for dishwasher tasks

**Hawai'i AI Initiative (HawAII), University of Hawai'i at Mānoa** — Honolulu, HI
*Student Researcher* (Advisor: Prof. Igor Molybog) — Mar 2026 – Sep 2026

**Fiatlux: Humanoid Ladder-Climbing and Light-Bulb Replacement Benchmark** | Project page

- Co-developed an **Isaac Lab** benchmark in which a **Unitree G1 humanoid** climbs a ladder to replace a bulb
- Split the task into **12 subtask environments** with difficulty-weighted scoring and domain randomization
- Released **RSL-RL PPO** and zero-shot **GR00T N1.7** VLA baselines; none completes a single subtask

**AI for Industry Challenge (Intrinsic & Open Robotics)** | GitHub

- Upgraded a privileged expert for **UR5e cable insertion** with force-monitored descent and jam recovery
- Built a **ROS 2/Gazebo** demo-collection pipeline on **Slurm** and a parallel **MCAP-to-LeRobot** converter
- Trained **ACT** and **Diffusion Policy** policies from expert demos with a **DAgger**-style approach

**Data Curation for Visual Language Reasoning**

- Formulated the research question of whether synthetic reasoning-step data improves **VQA** performance
- Built a **scalable data pipeline** turning images into question–answer pairs with intermediate reasoning
- Partnered with teammates to iterate on the research design and critique results

**Spinner Dolphin Photo Grading Protocol**

- Developed a **multi-stage model that detects, rates, and classifies** dolphins by fin distinctiveness
- Automated image-data pipelines, **significantly reducing manual processing time**
- Conducted **statistical analyses** to identify correlations between environmental factors and photo quality

## Work Experience

**Hyundai Card** — Seoul, South Korea
*Data Science Intern* — Jun 2025 – Jul 2025

- Built a **retrieval-augmented generation (RAG)** pipeline from scratch for text-to-SQL tasks
- Implemented **self-consistency decoding** and benchmarked LLM backbones and embedding models for accuracy
- Cut inference costs by moving from a closed- to an open-source model while preserving **80% accuracy**

**University of Hawai'i at Mānoa** — Honolulu, HI
*Software Verification Developer* — May 2024 – May 2025

- Applied formal methods to verify functional correctness in a project inspired by **seL4** and **RT-11**
- Designed and implemented a proof strategy using **interactive theorem proving (Isabelle/HOL)**

*Research Assistant* — Jun 2023 – Aug 2023

- Devised an efficient algorithm to simulate a mathematical model of biological reactions
- Created **2D and 3D models** to visualize the algorithm's results
- Collaborated with peers and professors to adapt the algorithm to additional considerations and scenarios

## Leadership

**Programming and Algorithms Club (PANDA)**
*Lecturer and Curriculum Developer* Aug 2022 – Aug 2025

- Earned the **2023 Best Officer (Big Panda) Award**
- Designed a specialized curriculum tailored to students' programming levels
- Led small-group lectures and discussions on developing and optimizing algorithms

**Association for Computing Machinery (ACM) at Mānoa**
*Server Administrator and Webmaster* Oct 2023 – Oct 2024

- Maintained and improved the chapter's software and official website using **React** and **Ruby/Jekyll**
- Built custom applications for chapter leadership on request
- Coordinated with a software engineering team via online collaboration platforms to develop and launch software

## Other Projects

**Jane Street Real-Time Market Data Forecasting** | GitHub

- Modeled complex market behavior with a **supervised autoencoder combined with an MLP**
- Engineered features to improve predictive accuracy and reflect real-world market conditions
- Ranked **440th of 3,000** participants (**top 15%**)

**BirdCLEF 2024** | GitHub

- Identified under-studied Indian bird species from call spectrograms using a **TensorFlow MobileNetV2** CNN
- Applied **transfer learning** to improve accuracy and AUROC

**Higgs Boson Detection 2024** | GitHub

- Classified events into those that produce the exotic **Higgs boson** particle and those that do not
- Benchmarked multiple **scikit-learn** models, analyzing the strengths and weaknesses of each
- Ranked **11th of 52** on the public leaderboard and **12th of 52** on the private leaderboard

**OpenAI CarRacing** | GitHub

- Developed an AI agent that drives on randomly generated race tracks
- Built a **2D convolutional neural network** with dropout to optimize the agent's performance
- Programmed the agent's training and inference phases in **TensorFlow**

**Full Cycle TakeOut Web App** | GitHub

- Named a **finalist** at the 2023 Hawai'i Annual Code Challenge (HACC)
- Created the app with support from the nonprofit ZeroWaste Oahu and the Full Cycle TakeOut program
- Implemented the app in two technology stacks: **MERN** and **Meteor**

**Protein Folding Algorithm** | GitHub

- Simulated the **hydrophobic-polar (HP) protein folding model** with recursive backtracking to find the optimal fold
- Split the project across a small, cross-functional team using a divide-and-conquer approach
- Delivered the research project under sponsorship from **Decision Research Corporation**